II. Curriculum Vitae

Personal Data

Title	Dr.
Name	Cong, Lei
Current position	Postdoc at Johannes Gutenberg-Universität Mainz (JGU)
Identifiers/ORCID	https://orcid.org/0000-0003-0002-1840
Research topic	Fundamental Physics, and Atomic, Molecular, Optical Physics (AMO)

Qualifications and Career

1. Professional experiences

Duration	Position	Supervisor:	Topic
07/03/2024-Present	Postdoc (HIM - JGU, Germany)	Dmitry Budker	Fundamental Physics and AMO
07/03/2022-06/03/2024	Postdoc (HIM - GSI, Germany)	Dmitry Budker	Fundamental Physics and AMO
24/10/2018-06/03/2022[#]	Postdoc (Shanghai University, China)	Enrique Solano	AMO

[#] Research visit: 05/07/2019–26/09/2020 — University of the Basque Country, Spain

2. Education (Ph.D. award date: 20.06.2018)

01/09/2013-20/06/2018	Ph.D.[+], Lanzhou University	Hong-gang Luo	Theoretical Physics
01/09/2009-20/06/2013	B. S., Lanzhou University	National Talent Training Base	Physics

[+] 5 years successive master-doctor program.

Oral Presentations at International Conferences:

1. [Invited] Excellence Custer "PRISMA++" Colloquium, Mainz, 2027
2. [Invited] Windows into New Physics in the Sky, Mainz MITP workshop, 2025
3. [Invited] Axions in Stockholm, Nordita, Stockholm, 2025
4. [Invited] 10th International Conference on Precision Physics and Fundamental Physical Constants (FFK2025), Warsaw, 2025
5. [Invited] International Symposium on New Physics Beyond the Standard Model, Sanya, 2025
6. [Selected] WE-Heraeus Seminar "Quest of New Physics at the Precision Frontier", Berlin, 2026
7. Fundamental constants and Physics beyond the Standard Model, ICAP, Hefei, 2026
8. APS Global Physics Summit (APS March Meeting), Los Angeles, 2018 & Denver, 2026
9. Quantum Sensing for Fundamental Physics, Mainz MITP workshop, 2024

10. Strongly Coupled Light-Matter Interactions, The Australian National University (ANU), 2020

Academic Service

1. Meeting:

- Organizer: Workshop on Spin-dependent exotic interactions and precision tests of fundamental symmetries (approved), ECT*, Italy, 2027.
- Organizer: Quantum Seminar for JGU Physics Department, Winter semester 2026/27.
- Session Chair for "Particles and Fields III" at APS Global Physics Summit 2026.
- Session Chair for "Q63 Strong light-matter interaction" DPG , Freiburg, 2024.
- Session Chair for "Search for new particles 6" at DPG conference, Karlsruhe, 2024.
- Committee member: 17th PATRAS workshop on axions, WIMPs, and WISPs held in Mainz, Germany, 2022.

2. Served as recording secretary (Protokollführer) for two Ph.D. defenses (JGU):

- A journey through nonlinear magneto-optics in Mainz, Germany, 2024.
- Searches for ultralight bosonic dark matter with atomic spectroscopy in Mainz, 2023.

3. External reviewer for DAAD (German Academic Exchange Service) funding proposals.

4. External reviewer for COST Action (European Cooperation in Science and Technology)

Academic Distinctions

1. Fellowships & Funding

✧ Co-Principal Investigator (Co-PI), DFG Individual Research Grant (Project No. 578071833)

✧ Helmholtz-OCPC (Office of the China Postdoctoral Council) Postdoc Fellowship
Project Title: Fundamental Physics, Atomic, Molecular, and Optical Physics
Funding amount: ~ 75,000 Euro
Duration: Mar., 2022 – Mar., 2024

✧ Postdoctoral Science Foundation Research Funding
Project Title: Quantum Simulation of Quantum Rabi Model in Trapped Ion System
Funding amount: ~ 11,000 Euro
Duration: Sep., 2019 – Sep., 2021

✧ National Natural Science Foundation of China (participate)
Project Title: Hybrid quantum system: the building block of quantum many-body problems
Funding amount: ~ 440,000 Euro (Duration: Sep., 2018 – Sep., 2022)

2. Excellent Graduate (Ph.D.), 2018

3. Merit Postgraduate, 2017

Supervision and Teaching

1. Supervised two PhD students, both of whom published first-author papers in Phys. Rev. A, and an undergraduate intern subsequently admitted to Princeton University with a full PhD scholarship.

2. Activity in teaching: Participated in teaching courses about “Basic Physics and Precision Measurements” (2024 & 2026) at JGU, Mainz.

Scientific Results

1. This review for Reviews of Modern Physics addresses a fundamental topic in modern physics: spin-dependent exotic interactions mediated by hypothetical bosons such as axions, which are promising candidates for dark matter and may help resolve other open problems in fundamental physics. This three-year project demonstrates my ability to lead and deliver a comprehensive international research effort.

- ✧ **L. Cong*,** W. Ji*, P. Fadeev, F. Ficek, M. Jiang, V. V. Flambaum, H.S. Guan, D. F. Jackson Kimball, M. G. Kozlov,Y. V. Stadnik, D. Budker.

  Title: Spin-dependent exotic interactions

  Journal: Rev. Mod. Phys. **97**, 025005 (2025). (Citation: 97)

  URL: https://doi.org/10.1103/RevModPhys.97.025005

* Corresponding author

2. The following papers demonstrate the use of precision platforms, from AMO systems to levitated magnets, in searches for new physics beyond the Standard Model.

- ✧ **K. Gaul*, L. Cong*, D. Budker**

  Title: Constraints on New Vector Boson Mediated Electron-Nucleus Interactions from Spectroscopy Data of Polar Diatomic Molecules

  Journal: *Phys. Rev. Lett.* **136**, 181805 (2026)

  URL: https://doi.org/10.1103/d19m-s856

- ✧ Z.T. Xu, X. Heng, G.Q. Tian, D. Gong, **L. Cong**, W. Ji*, D. Budker, K. Wei*

  Title: Constraints on Axion-Mediated Dipole-Dipole Interactions

  Journal: *Phys. Rev. Lett.* **134**, 181801(2025)

  URL: https://doi.org/10.1103/PhysRevLett.134.181801

- ✧ Y.H. Wang, X.M. Huang, J.X. Li, W.Z. Wang, X. Kang,Y.X. Liu, **L. Cong**, D. Budker, M. Jiang*, and X.H. Peng*

  Title: Magnetic-resonance searches for Earth-sourced exotic nuclear-spin-dependent interactions

  Journal: *Sci. Adv.* 12, eaeg0974 (2026)

- ✧ **L. Cong**, F.Ficek, P. Fadeev, D. Budker*

  Title: Improved constraints on exotic interactions between electron and proton in hydrogen

  Journal: *Phys. Rev. A* **112**, 052824 (2025)

  URL: https://doi.org/10.1103/8nyk-sysy

- ✧ D. W. P. Amaral*, **L. Cong***, T. M. Fuchs, H. Ulbricht,, and D. Budker

  Title: A Levitated-Magnet Vector Force Sensor for Spin-Dependent Exotic Interactions

  Journal: Submitted to *Phys. Rev. Lett.* (under review)

  URL: arXiv: 2606.30781

- ✧ **L. Cong**, Filip Ficek*, P. Fadeev, Y. V. Stadnik, D. Budker

  Title: Searching for Exotic Interactions between Antimatter

  URL: Submitted to Science Advances (under review), arXiv:2503.07161

- ✧ **L. Cong**, F. Ficek*, R.Abdullin, M.G. Kozlov*, D. Budker

  Title: Testing Exotic Electron-Electron Interactions with the Helium Ionization-Energy Anomaly

  URL: *Phys. Rev. D* **114** , 095010 (2026)

- ✧ **L. Cong**, D. F. Jackson Kimball, M. G. Kozlov, and D. Budker*

  Title: Constraints on exotic interactions from scalar spin-spin coupling in tritium duteride

  URL: arXiv:2408.15442

3. I have a solid background in quantum optics, especially in modeling light–matter interaction systems. The following publications demonstrate my experience with quantum-optical models as well as their implementation in real systems such as trapped-ion platforms.

- ✧ **L. Cong**, S. Felicetti, J. Casanova, L. Lamata, E. Solano* and I. Arrazola*.

  Title: Selective Interactions in the Quantum Rabi Model.

  Journal: *Phys. Rev. A* **101**, 032350 (2020). (Citation: 43)

  URL: *https://doi.org/10.1103/PhysRevA.101.032350*

- ✧ **L. Cong**, X.-M. Sun, M. Liu, Z.-J. Ying*, and H.-G. Luo*.

  Title: Polarons picture in the two-photon quantum Rabi model.

  Journal: *Phys. Rev. A* **99**, 013815 (2019). (Citation: 84)

URL： *https://doi.org/10.1103/PhysRevA.99.013815*

✧ **L. Cong**, X.-M. Sun, M. Liu, Z.-J. Ying*, and H.-G. Luo*.

Title: Frequency-renormalized multipolaron expansion for the quantum Rabi model.

Journal: *Phys. Rev. A.* **95**, 063803 (2017). (Citation: 41)

URL： *https://doi.org/10.1103/PhysRevA.95.063803*

✧ D. Braak*, **L. Cong***, H.-P. Eckle, H. Johannesson, E.-K. Twyeffort

Title: Spectral continuum in the Rabi–Stark model

Journal: *J. Opt. Soc. Am. B*, 41(8), C97-C111 (2024). (Citation: 11)

URL: *https://doi.org/10.1364/JOSAB.524014*

✧ Y. Shi, **L. Cong***, and H.-P. Eckle*.

Title: Entanglement resonance in the asymmetric quantum Rabi model.

Journal: *Phys. Rev. A* **105**, 062450 (2022). (Citation: 29)

URL: *https://doi.org/10.1103/PhysRevA.105.062450*

✧ X.-M. Sun, **L. Cong**, H.-P. Eckle, Z.-J. Ying* and H.-G. Luo*.

Title: Application of polaron picture in the two-qubit quantum Rabi model.

Journal: *Phys. Rev. A* 101, 063832 (2020). (Citation: 16)

URL： *https://doi.org/10.1103/PhysRevA.101.063832*

4. My development of a community webpage, contributions to a COST Action working-group white paper, and invited contribution to the development of a CERN DRD5 (Detector R&D Collaboration 5: Quantum Sensors for Particle Physics) COST Action proposal demonstrate my active contribution to the broader community developing sensing technologies for fundamental physics.

✧ Webpage for spin-dependent fifth force (1,450 visits from more than 11 countries since 2025)

URL: https://lei-cong.github.io/Spin-Dependent-5th-Force-Limits/

✧ A. Arza, et *al.,* **L. Cong,** *et al.*

Title: The COSMIC WISPers White Paper: The physics case for Weakly Interacting Slim Particles

Journal: arXiv:2603.03433 (Submitted to Eur. Phys. J. D)

✧ Georgy Kornakov(The Main Proposer), **L. Cong** (Secondary Proposer) *et al.*

"Bridging Research and Application of Quantum Sensors"

- ✧ 2026–Present | Member, ESA Topical Team “ACES and General Relativity” European Space Agency (ESA)